# Software Requirements Specification (SRS)

**Document Version:** 1.0
**Project Name:** Web-Based To-Do Application
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## 1. Introduction

### 1.1 Purpose

The purpose of this document is to define the requirements for a web-based To-Do application that allows users to create, manage, and organize todos efficiently. This document will serve as a guide for developers, designers, testers, and stakeholders during the development process.

### 1.2 Scope

The Web-Based To-Do Application will provide users with a simple and intuitive interface to:

- Create, update, and delete todos.
- Mark todos as completed.
- Access todos from any device with a web browser.

The application will be accessible via a web browser and will not require any installation.

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## 2. Overall Description

### 2.1 Product Perspective

The To-Do application is a standalone web application that will be hosted on a cloud server. It will be accessible via modern web browsers (e.g., Chrome, Firefox, Safari, Edge).

### 2.2 User Classes

- **End Users:** Individuals who will use the application to manage their todos.

### 2.3 Operating Environment

- **Frontend:** HTML, CSS, JavaScript Next.js or Vue.js recommended).
- **Backend:** Node.js or Next.js.
- **Database:** PostgreSQL, MySQL, or MongoDB.
- **Hosting:** Containerized on cloud-based hosting (e.g., AWS, Heroku, or DigitalOcean) behind an nginx proxy manager.

### 2.4 Design and Implementation Constraints

- The application must be responsive and work on both desktop and mobile devices.
- The application must support modern web browsers.
- Data must be securely stored and transmitted (HTTPS).

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## 3. System Features

### 3.1 User Authentication

- Users can create an account using an email address and password.
- Password must be between 3  and 10 characters in length.
- Users can log in and log out of their accounts.
- Password reset functionality will be available.

### 3.2 Todo Management

- Users can create new todos with the following attributes:
    - Title (required)
- Users can edit or delete existing todos.

### 3.4 User Interface

- The interface will be clean, intuitive, and responsive.
- Todos will be displayed in a list.

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## 4. Functional Requirements

| ID | Requirement | Description |
|---|---|---|
| FR-1 | User Registration | Users can create an account. |
| FR-2 | User Login/Logout | Users can log in and log out of their accounts. |
| FR-3 | Create Todo | Users can create a new todo with a title, description, due date, and priority. |
| FR-4 | Edit Todo | Users can edit an existing todo. |
| FR-5 | Delete Todo | Users can delete a todo. |
| FR-6 | Mark Todo as Completed | Users can mark a todo as completed. |
| FR-7 | Filter Todos | Users can filter todos by due date, priority, or completion status. |
| FR-8 | Search Todos | Users can search for todos by title or description. |
| FR-9 | Set Reminders | Users can set reminders for todos with due dates. |
| FR-10 | Organize Todos | Users can organize todos into categories or projects. |

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## 5. Non-Functional Requirements

| ID | Requirement | Description |
|---|---|---|
| NFR-1 | Performance | The application should load in under 3 seconds. |
| NFR-2 | Scalability | The application should support up to 10,000 concurrent users. |
| NFR-3 | Security | User data must be encrypted during transmission and storage. |
| NFR-4 | Availability | The application should have 99.9% uptime. |
| NFR-5 | Usability | The application should be intuitive and easy to use. |
| NFR-6 | Compatibility | The application should work on all major web browsers. |

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## 6. External Interface Requirements

### 6.1 User Interfaces

- **Home Page:** Displays a list of todos and options to create new todos.
- **Todo Details Page:** Displays detailed information about a todo.
- **Login/Register Page:** Allows users to log in or create an account.

### 6.2 Hardware Interfaces

- The application will run on standard web servers and cloud hosting platforms.

### 6.3 Software Interfaces

- **Frontend:** HTML, CSS, JavaScript.
- **Backend:** RESTful API for communication between frontend and backend.
- **Database:** SQL or NoSQL database for storing user data and todos.

### 6.4 Communication Interfaces

- The application will use HTTPS for secure communication.

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## 7. Other Non-Functional Requirements

### 7.1 Performance Requirements

- The application should handle up to 10,000 concurrent users without performance degradation.

### 7.2 Security Requirements

- User passwords must be hashed and salted before storage.
- The application must use HTTPS to encrypt data in transit.

### 7.3 Maintainability

- The codebase should be well-documented and follow best practices for readability and scalability.

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## 8. Appendices

### 8.1 Glossary

- **Todo:** A single to-do item with a title, description, due date, and priority.
- **Category/Project:** A group of related todos.
- **Reminder:** A notification to alert the user about an upcoming todo.

### 8.2 References


- https://www.blazemeter.com/blog/selenium-github


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This SRS document provides a comprehensive overview of the requirements for the Web-Based To-Do Application. It can be further refined based on stakeholder feedback and project-specific needs.

# Sprint Breakdown for the Web-Based To-Do Application

## Sprint 0: Preparation

**Duration:** 1 week
**Objective:** Set up the project, define requirements, and prepare the testing environment.

**Activities:**

1. **Requirement Gathering:**
    - Define user stories and acceptance criteria.
    - Prioritize features for the MVP.
2. **Test Planning:**
    - Create a high-level test plan.
    - Identify testing tools and frameworks.
3. **Environment Setup:**
    - Set up development, and test environments.
    - Configure CI/CD pipelines (e.g., Jenkins, GitHub Actions).
4. **Test Case Design:**
    - Draft test cases for core features (e.g., user registration, todo creation).

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## Sprint 1: User Authentication

**Duration:** 2 weeks
**Objective:** Implement and test user registration and login functionality.

**Features Delivered:**

- User registration (email and password).
- User login and logout.
- Password recovery.

**Testing Activities:**

1. **Unit Testing:**

    - Test backend logic for user registration and authentication.
2. **Integration Testing:**
    - Test API endpoints for user registration and login.
3. **System Testing:**
    - Test end-to-end workflows for registration, login, and password recovery.
4. **Defect Reporting:**
    - Log and prioritize defects in the tracking tool.

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## Sprint 2: Todo Management

**Duration:** 2 weeks
**Objective:** Implement and test core todo management features.

**Features Delivered:**

- Create, edit, and delete todos.
- Mark todos as completed.

**Testing Activities:**

1. **Unit Testing:**
    - Test backend logic for todo creation, editing, and deletion.
2. **Integration Testing:**
    - Test API endpoints for todo management.
3. **System Testing:**
    - Test end-to-end workflows for todo creation, editing, and deletion.
4. **Regression Testing:**
    - Ensure user authentication features are not broken.

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## Sprint 3: User Interface and Usability

**Duration:** 2 weeks
**Objective:** Finalize the user interface and ensure usability.

**Features Delivered:**

- Clean and intuitive UI design.
- Responsive design for desktop and mobile devices.
- Light and dark theme support.

**Testing Activities:**

1. **UI Testing:**
    - Test the application on different devices and browsers.
2. **Usability Testing:**
    - Conduct usability tests with real users.
3. **Accessibility Testing:**
    - Ensure the application is accessible to users with disabilities.
4. **Regression Testing:**
    - Ensure all previous features are not broken.

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## Sprint 4: Final Testing and Release

**Duration:** 2 weeks
**Objective:** Conduct final testing and prepare for release.

**Activities:**

1. **User Acceptance Testing (UAT):**
    - Validate the application against business requirements.
2. **Performance Testing:**
    - Test the application under load to ensure it meets performance requirements.
3. **Security Testing:**

    - Verify that user data is securely stored and transmitted.
4. **Bug Fixing:**
    - Address any critical defects identified during UAT.
5. **Release Preparation:**
    - Prepare release notes and documentation.
    - Deploy the application to the production environment.

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## Summary of Sprints

| Sprint | Focus Area | Key Deliverables |
|---|---|---|
| Sprint 0 | Preparation | Test plan, environment setup, test case design. |
| Sprint 1 | User Authentication | User registration, login, password recovery. |
| Sprint 2 | Todo Management | Create, edit, delete, and mark todos as completed. |
| Sprint 3 | Todo Organization and Filtering | Organize todos, filter todos, search todos. |
| Sprint 4 | Notifications and Reminders | Set reminders, display notifications, send email reminders. |
| Sprint 5 | User Interface and Usability | Clean and responsive UI, light/dark themes. |
| Sprint 6 | Final Testing and Release | UAT, performance testing, security testing, bug fixing, and release preparation. |

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This sprint breakdown ensures that the **Web-Based To-Do Application** is developed and tested incrementally, with a focus on delivering value to the end-user in each sprint. Let me know if you need further adjustments or details!

# User stories

## Sprint 0: Preparation

**User Stories and Test Cases with Acceptance Criteria:**

1. **As a product owner, I want to define user stories and acceptance criteria so that development aligns with business needs.**
    - **Acceptance Criteria:**
        - All user stories must cover core features.
        - Acceptance criteria must be clearly defined and agreed upon.
    - **Test Cases:**
        - Verify that user stories and acceptance criteria cover all key features.
        - Ensure acceptance criteria align with business requirements.
2. **As a tester, I want to create a high-level test plan so that testing activities are well-structured.**
    - **Acceptance Criteria:**
        - Test plan includes unit, integration, system, and regression testing.
    - **Test Cases:**
        - Validate that the test plan includes all necessary testing phases.
        - Ensure traceability between test cases and requirements.

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## Sprint 1: User Authentication

1. **As a user, I want to register with my email and password so that I can create an account.**
    - **Acceptance Criteria:**
        - User must provide a valid email and password.
        - Passwords must be securely stored.
    - **Test Cases:**

        - Verify registration with a valid email and password.
        - Validate error messages for invalid email formats.
        - Ensure password hashing and storage security.
2. **As a user, I want to log in using my credentials so that I can access my todos.**
    - **Acceptance Criteria:**
        - Login should be successful with correct credentials.
        - Incorrect credentials should trigger an appropriate error message.
    - **Test Cases:**
        - Test successful login with correct credentials.
        - Verify login fails with incorrect credentials.
3. **As a user, I want to recover my password so that I can regain access if I forget it.**
    - **Acceptance Criteria:**
        - User should receive a password reset email when requested.
        - Password reset should only work for registered emails.
    - **Test Cases:**
        - Test password recovery flow, including email verification.
        - Validate security of the reset link.

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## Sprint 2: Todo Management

1. **As a user, I want to create a new todo item so that I can track tasks.**
    - **Acceptance Criteria:**
        - User must provide a title for the todo.
        - Todo should be saved successfully.
    - **Test Cases:**
        - Verify that a new todo can be created with a title and description.
        - Ensure todos are persisted correctly.

2. **As a user, I want to edit a todo item so that I can update task details.**
    - **Acceptance Criteria:**
        - User must be able to modify title and description.
    - **Test Cases:**
        - Ensure todos can be edited and updated correctly.
3. **As a user, I want to delete a todo item so that I can remove tasks I no longer need.**
    - **Acceptance Criteria:**
        - Deleted todos should not be retrievable.
    - **Test Cases:**
        - Validate that deleting a todo removes it permanently.

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## Sprint 3: User Interface and Usability

1. **As a user, I want a clean and intuitive UI so that I can navigate easily.**
    - **Acceptance Criteria:**
        - UI should be user-friendly and accessible.
    - **Test Cases:**
        - Verify UI elements are correctly aligned and accessible.
2. **As a user, I want the application to be responsive so that it works on both desktop and mobile devices.**
    - **Acceptance Criteria:**
        - Layout should adjust based on device type.
    - **Test Cases:**
        - Test responsiveness across different screen sizes and browsers.

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## Sprint 4: Final Testing and Release

1. **As a business owner, I want to validate the application against business requirements so that it meets expectations.**
    - **Acceptance Criteria:**
        - Application should function as per initial requirements.
    - **Test Cases:**
        - Conduct user acceptance testing to validate functionality.
2. **As a security analyst, I want to verify user data security so that user information is protected.**
    - **Acceptance Criteria:**
        - Sensitive data should be encrypted.
    - **Test Cases:**
        - Verify security measures, including encryption and authentication mechanisms.

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This structure ensures each sprint has clear user stories with well-defined acceptance criteria and corresponding test cases.

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# Contents

Software Requirements Specification (SRS) ..... 1
1. Introduction ..... 1
2. Overall Description ..... 1
3. System Features ..... 2
4. Functional Requirements ..... 3
5. Non-Functional Requirements ..... 3
6. External Interface Requirements ..... 4
7. Other Non-Functional Requirements ..... 4
8. Appendices ..... 5
Sprint Breakdown for the Web-Based To-Do Application ..... 6
Sprint 0: Preparation ..... 6
Sprint 1: User Authentication ..... 6
Sprint 2: Todo Management ..... 7
Sprint 3: User Interface and Usability ..... 8
Sprint 4: Final Testing and Release ..... 8
Summary of Sprints ..... 9
User stories ..... 10
Sprint 0: Preparation ..... 10
Sprint 1: User Authentication ..... 10
Sprint 2: Todo Management ..... 11
Sprint 3: User Interface and Usability ..... 12
Sprint 4: Final Testing and Release ..... 13
Team members ..... 14